

EFFICIENT BAYESIAN INFERENCE FROM NOISY PAIRWISE COMPARISONS

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ABSTRACT

Evaluating generative models is difficult because automatic metrics correlate weakly with human preference judgments. Human evaluation is more faithful but incurs high annotation cost and exhibits substantial label noise, since annotators vary in expertise, attention, and diligence. Pairwise comparison protocols reduce response variance relative to absolute scoring, but recovering a global quality ordering from a sparse set of comparisons requires an explicit aggregation model. Bradley-Terry estimators map comparison outcomes to per-item latent scores; however, existing formulations either assume homogeneous annotator reliability or optimize objectives that admit no convergence guarantee, which limits both robustness and parameter interpretability. We introduce BBQ, a Bayesian Bradley-Terry model that augments the item-skill parameters with an explicit per-rater reliability parameter, thereby down-weighting or excluding unreliable annotators, and that admits a provably monotone increase of the observed-data likelihood under an Expectation-Maximization (EM) procedure with closed-form updates. Empirically, BBQ converges in fewer iterations, yields calibrated posterior uncertainty over item scores, and produces rankings that are more robust and more interpretable than baseline Bradley-Terry estimators, including under noisy crowdsourced annotation. Taken together, these properties make BBQ a more reliable and cost-efficient instrument for the human evaluation of generative models.

1 INTRODUCTION

Evaluating generative models is difficult, particularly for large language models (LLMs) and image generators, because automatic metrics correlate weakly with human perceptual preferences. Reference-based and distributional metrics such as BLEU (Papineni et al., 2002) and perplexity (Jelinek, 1998) for LLMs, or PSNR (Gonzalez, 2009), MS-SSIM (Wang et al., 2003), and FID (Heusel et al., 2017) for image models, capture only surface statistics and are weak proxies for perceived quality (Mentzer et al., 2020; CLIC, 2025; Chiang et al., 2024). Consequently, human judgments remain the reference standard for establishing quality orderings across models. For this reason, leaderboards and comparative studies of LLMs and Learned Image Compression (LIC) methods are constructed primarily from elicited human preference data.

Eliciting human judgments, however, is costly in both annotator time and budget, which makes it essential to adopt elicitation protocols that maximize information per annotation while controlling subjectivity and label noise. Within this design space, pairwise comparison, in which an annotator selects the preferred item from a presented pair rather than reporting an absolute score, is an effective and low-variance elicitation format (Zerman et al., 2018; Wang et al., 2023). The forced-choice format imposes a lower cognitive load on annotators and yields judgments with reduced response variance. Exhaustive collection of all pairs is intractable, since the number of distinct pairs scales as $O(K^2)$ in the number of items K . Under a sparse comparison budget, marginal statistics such as empirical win rates are inadequate estimators of global quality, because a win rate is confounded by the skill of the opponents against which an item was sampled.

Recovering a global ranking suitable for a leaderboard therefore requires an aggregation model that maps the observed comparison outcomes to a consistent set of item scores. This requirement motivates aggregation estimators that are both statistically robust and computationally efficient. The Bradley-Terry (BT) model (Bradley & Terry, 1952) formalizes this task as maximum-likelihood

estimation of latent item scores from binary comparison outcomes, optimized by iterative updates. It is the de facto standard estimator for aggregating pairwise judgments across application domains.

Despite its wide adoption, the standard Bradley–Terry estimator exhibits two limitations that are material in practice. First, its parameters may be estimated by several distinct optimization schemes, including iterative scaling (Ford Jr, 1957), first-order gradient ascent, and the minorization-maximization (MM) algorithm (Hunter, 2004). For the basic BT model the log-likelihood is concave, and MM is guaranteed to converge to the global maximizer; gradient-based updates, which are algorithmically equivalent to the Elo rating recursion, are prevalent in practice but carry no such convergence guarantee (Hunter, 2004). For extended BT models that incorporate additional structure, such as home-field advantage (Agresti, 2010), multi-item comparisons (Plackett, 1975; Luce et al., 1959), or ties (Rao & Kupper, 1967), the objective is no longer concave, and even MM is only guaranteed to reach a local optimum (Hunter, 2004). Second, human-elicited comparisons are heteroscedastic in reliability: annotators may respond hastily, guess, or attend inconsistently, so the label-noise level varies across raters. When this per-rater reliability is treated as homogeneous, low-reliability annotators bias the estimated item scores and degrade the stability of the recovered ranking.

We propose a Bayesian Bradley–Terry model that jointly infers the latent item scores and per-rater reliability parameters. To our knowledge, no prior work couples the BT likelihood with a Bayesian treatment that parameterizes rater quality explicitly while admitting closed-form EM updates. Prior methods that model annotator noise typically resort to gradient-based optimization without convergence guarantees; by contrast, our estimator attains a monotone likelihood ascent and yields directly interpretable parameters. Placing conjugate priors on the parameters quantifies epistemic uncertainty and regularizes the estimates. To capture heterogeneity in annotator behavior, we introduce a per-rater reliability parameter that quantifies each annotator’s consistency, and that rescales the contribution of that annotator’s comparisons to the likelihood. From a latent-variable formulation of the mixture likelihood, we derive an EM algorithm that estimates the item and rater parameters at low per-iteration cost while guaranteeing a monotone increase of the observed-data likelihood. This mechanism retains the informative signal from partially reliable annotators while attenuating the contribution of inattentive or inconsistent ones. Because the estimator is Bayesian and models rater quality explicitly, it also returns posterior uncertainty over item scores, which supports interpretable rankings and cross-study comparisons.

We evaluate the estimator on several human-evaluation datasets and additionally introduce a newly collected *inhomogeneous rater quality (IHQ)* dataset¹. Relative to the standard BT estimator and to naive marginal aggregation, our method converges in fewer iterations, is more robust to comparison noise, and recovers more consistent rankings. Beyond generative-model evaluation, the formulation applies to any setting in which noisy pairwise comparisons must be aggregated into a reliable global ranking. In sum, this work contributes methodology for more cost-efficient, interpretable, and reproducible human studies of AI-generated content.

2 RELATED WORK

Many human studies reduce to the problem of ranking a set of items. A two-alternative forced-choice format, in which the annotator selects between two items rather than assigning an absolute score to a single item, increases discriminative sensitivity to small quality differences and lowers response variance (Zerman et al., 2018; Wang et al., 2023). For this reason, many machine-learning leaderboards are built on pairwise comparison data. In such protocols, each annotator reports a preference over a presented pair, and the collected preferences are subsequently aggregated into a global ranking. Representative instances are the Chatbot Arena (Chiang et al., 2024) and the CLIC image-compression challenge (CLIC, 2025), both of which elicit pairwise preferences and aggregate them with variants of the Bradley–Terry (BT) model.

The Bradley–Terry model (Bradley & Terry, 1952) was introduced to rank competitors from paired outcomes in sports. It specifies a probabilistic model for the event that one item is preferred over another as a function of the two items’ latent scores. By mapping pairwise outcomes to a total ordering, it has become a foundational estimator across games, consumer-preference elicitation, and

¹The IHQ dataset will be released upon acceptance.

related applications. A number of extensions generalize its modeling assumptions. The Plackett-Luce model, for instance, extends the BT likelihood from pairwise comparisons to full rankings over multiple items (Plackett, 1975; Luce et al., 1959), inducing a distribution over permutations as a product of successive BT choice probabilities (Luce et al., 1959). Other variants address specific observation models, including home-field advantage (Agresti, 2010), the presence of ties (Rao & Kupper, 1967), and comparisons between groups rather than individual items (Huang et al., 2006).

The parameters of the basic BT model are typically inferred by maximum-likelihood estimation (MLE). Zermelo (1929) proposed an iterative fixed-point scheme for computing the MLE that remains in wide use. Lange et al. (2000) subsequently showed that this scheme is an instance of the minorization-maximization (MM) family, which iteratively maximizes a tangent surrogate lower bound and converges to a local maximum of the likelihood. Hunter (2004) generalized MM to extended BT models and derived sufficient conditions under which the iterates converge to the MLE. For the classical BT model the log-likelihood is concave, so MM converges to the global optimum (Hunter, 2004). For the extensions above, home-field advantage, multi-item comparisons, or ties, concavity no longer holds, and MM is only guaranteed to converge to a local maximum (Hunter, 2004).

Bayesian formulations of the BT model incorporate prior information and act as regularizers (Adams, 2005; Guiver & Snelson, 2009; Caron & Doucet, 2012). Caron & Doucet (2012) established that the MM iteration admits an Expectation-Maximization (EM) interpretation under a latent-variable augmentation, which in turn enables Bayesian inference via Gibbs sampling. This augmentation yields a tractable complete-data likelihood and guarantees convergence of the associated Markov Chain Monte Carlo sampler. The Bayesian treatment shrinks the posterior estimates, mitigates sensitivity to local maxima, and furnishes posterior uncertainty over the estimated item skills.

Annotator-quality models relax the homogeneous-reliability assumption that all comparisons carry equal evidential weight. In crowdsourced elicitation, per-annotator reliability varies substantially, which motivates likelihoods that parameterize annotator reliability explicitly (Chen et al., 2013). Chen et al. (2013) introduced a Bayesian model that jointly estimates item scores and per-annotator reliability through annotator-specific parameters. Their formulation places Gaussian priors on the item and rater parameters and introduces per-annotator scaling factors in the likelihood that modulate each annotator’s influence. They perform posterior inference by gradient-based optimization, which is difficult in the high-dimensional parameter space owing to non-convexity and local optima, and which offers no convergence guarantee. Our EM-based estimator, by contrast, is guaranteed to increase the observed-data likelihood monotonically at every iteration. A simplified Elo-based rater model developed by Google Research instantiates a heuristic variant of this idea. It is used in the Challenge on Learned Image Compression (CLIC) (CLIC, 2025) and in related work (Mentzer et al., 2020; Ballé et al., 2025).

To our knowledge, no prior work combines the Bradley–Terry likelihood with a Bayesian formulation that parameterizes rater quality explicitly and simultaneously admits closed-form EM updates. Unlike Chen et al. (2013), who adopt Gaussian priors and gradient-based optimization without convergence guarantees, our formulation uses conjugate priors that render the M-step analytic and yields an EM algorithm with a monotone likelihood ascent. Relative to the Elo-based rater model used in CLIC (CLIC, 2025), which is a heuristic built on Elo-style incremental updates, our estimator provides a principled Bayesian treatment with posterior uncertainty and interpretable rater-quality parameters.

3 METHODOLOGY

3.1 BRADLEY-TERRY MODEL WITH RATER QUALITY

Aggregating a set of noisy pairwise comparisons into a reliable item ranking is a fundamental estimation problem in machine learning and statistics. Let K items be compared pairwise, with replacement, by a set of R raters, and let D denote the observed data, i.e., the collection of realized comparison outcomes. For an ordered pair of items i and j , Bradley & Terry (1952) specify the

choice model

$$P(i \text{ beats } j) = \frac{\lambda_i}{\lambda_i + \lambda_j} \quad (1)$$

where $\lambda_k > 0$ is the latent skill parameter of item $k \in \{1, 2, \dots, K\}$, and the choice probability depends only on the two competitors' skills. This model yields an interpretable estimator of item scores from observed win/loss counts. Its likelihood, however, rests on the assumption that every comparison is equally reliable, i.e., that label noise is homogeneous across raters. In human evaluation this assumption is routinely violated: annotators differ in expertise, attention, and diligence, so their judgments differ in reliability.

To model heterogeneous reliability, we augment the likelihood with a rater-specific quality parameter $q_r \in [0, 1]$. Under this model, rater r produces an informed judgment drawn from the Bradley-Terry choice model with probability q_r , and with the complementary probability $1 - q_r$ responds uninformatively, choosing between the two items by a fair Bernoulli draw. Marginalizing over this two-component mixture gives the probability that rater r ranks item i above item j :

$$P(r \text{ ranks } i \text{ above } j) = q_r \left(\frac{\lambda_i}{\lambda_i + \lambda_j} \right) + (1 - q_r) \left(\frac{1}{2} \right) \quad (2)$$

The log-likelihood of the data is given by:

$$\begin{aligned} \log P(D \mid \lambda, q) = & \sum_{r=1}^R \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K \left[w_{r,ij} \log \left(q_r \frac{\lambda_i}{\lambda_i + \lambda_j} + (1 - q_r) \frac{1}{2} \right) \right. \\ & \left. + (n_{r,ij} - w_{r,ij}) \log \left(q_r \frac{\lambda_j}{\lambda_i + \lambda_j} + (1 - q_r) \frac{1}{2} \right) \right], \end{aligned} \quad (3)$$

where $n_{r,ij}$ is the total number of comparisons of items i and j by rater r , and $w_{r,ij}$ is the count of those comparisons in which rater r ranked item i above item j . This log-likelihood admits no closed-form maximizer, so the parameters λ and q cannot be obtained analytically. Gradient ascent is applicable but requires learning-rate tuning and provides no convergence guarantee. The Expectation-Maximization (EM) algorithm, by contrast, requires no step-size tuning and is guaranteed to increase the observed-data likelihood monotonically and to converge to a stationary point (Dempster et al., 1977).

3.2 THURSTONIAN INTERPRETATION

Following Caron & Doucet (2012), we recast the Bradley-Terry model in the Thurstonian random-utility framework (Diaconis, 1988). In this formulation, a comparison of items i and j is modeled as a race in which each item is assigned an independent latent arrival time, Y_i and Y_j . The arrival times are taken to be exponentially distributed with rates equal to the item skills,

$$Y_i \sim \mathcal{E}(\lambda_i), \quad Y_j \sim \mathcal{E}(\lambda_j), \quad (4)$$

and the item attaining the smaller arrival time is declared the winner. Because the minimum of independent exponentials is exponential and the probability that Y_i is smallest is proportional to its rate, this construction recovers the standard Bradley-Terry choice probability:

$$P(i \text{ beats } j) = P(Y_i < Y_j) = \frac{\lambda_i}{\lambda_i + \lambda_j}. \quad (5)$$

To render the complete-data likelihood tractable for EM, we augment the model with latent variables. First, we introduce the mixture-assignment indicator

$$A_{r,ij}^{(c)} \sim \text{Bernoulli}(q_r), \quad (6)$$

which encodes whether the c -th comparison of items i and j by rater r was drawn from the Bradley-Terry component ($A = 1$) rather than the uninformative component ($A = 0$). Conditioning on these

indicators, we define the latent statistic $Z_{r,ij}$ as the sum of the winning (minimal) arrival times over the $n_{r,ij}$ comparisons by rater r :

$$Z_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)} \min(Y_i^{(c)}, Y_j^{(c)}). \quad (7)$$

Conditioned on the number of Bradley-Terry-component comparisons $m_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)}$, and using that each winning arrival time is exponential with rate $\lambda_i + \lambda_j$, the statistic $Z_{r,ij}$ is Gamma-distributed,

$$Z_{r,ij} \mid m_{r,ij} \sim \Gamma(m_{r,ij}, \lambda_i + \lambda_j), \quad (8)$$

where $\Gamma(\alpha, \beta)$ denotes the Gamma distribution with shape α and inverse scale (rate) β . This Gamma-distributed latent statistic makes the complete-data likelihood conjugate in λ , which is precisely what yields closed-form EM updates while retaining the rater-specific quality parameters.

3.3 EXPECTATION-MAXIMIZATION UPDATES

We use EM to compute maximum a posteriori (MAP) estimates of the parameters λ and q , treating the mixture assignments A and the unobserved arrival times of the Thurstonian construction as latent variables. Under the standard EM guarantee, the iterates converge to a stationary point of the log-posterior.

We first specify conjugate priors on the parameters. Each item skill λ_k is assigned a Gamma prior, $\lambda_k \sim \Gamma(a, b)$, conjugate to the exponential arrival-time likelihood, and each rater-quality parameter q_r is assigned a Beta prior, $q_r \sim B(\alpha, \beta)$, conjugate to the Bernoulli mixture indicator.

Each iteration maximizes the expected complete-data log-posterior evaluated at the current estimates (λ^*, q^*) . The algorithm alternates the following two steps until the convergence criterion is met:

E-step: Expectation In the E-step we form the expected complete-data log-posterior, denoted Q : the expectation of the joint log-posterior over observed and latent variables, taken with respect to the posterior over the latent variables induced by the observed data D and the current estimates. It is defined as

$$Q(\lambda, q \mid \lambda^*, q^*) = \mathbb{E}_{A, Z \mid D, \lambda^*, q^*} [\ell_c(\lambda, q; D, Z, A) + \log P(\lambda) + \log P(q)]. \quad (9)$$

The complete-data log-likelihood ℓ_c factorizes into three additive terms:

$$\ell_c(\lambda, q; D, Z, A) = \log P(Z \mid D, A, \lambda, q) + \log P(A \mid D, \lambda, q) + \log P(D \mid \lambda, q). \quad (10)$$

M-step: Maximization The M-step updates the parameters by maximizing Q with respect to (λ, q) . Substituting the expected sufficient statistics computed in the E-step reduces the otherwise intractable joint optimization to a pair of closed-form updates. The quantity common to both updates is the posterior responsibility that a comparison from rater r was generated by the Bradley-Terry component rather than by chance. This responsibility, denoted γ , is the soft weight assigned to a judgment according to its consistency with the current model estimates, and is given by

$$\gamma_{r,ij}^{(t-1)} = \frac{q_r^{(t-1)} y_{ij}^{(t-1)}}{q_r^{(t-1)} y_{ij}^{(t-1)} + (1 - q_r^{(t-1)}) \frac{1}{2}}, \quad (11)$$

where $y_{ij}^{(t-1)} = \frac{\lambda_i^{(t-1)}}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}}$ is the Bradley-Terry probability that item i beats item j under the current estimates. The responsibility $\gamma \in [0, 1]$ is the posterior probability, given the current estimates, that a comparison was informative rather than random. Values of γ near one identify comparisons that the model treats as reliable.

The update for rater quality q_r is a regularized fraction of responsibility-weighted comparisons. The numerator accumulates rater r 's responsibility-weighted judgments, each contribution scaled by the posterior probability γ that the judgment was informative, and is shifted by the Beta-prior pseudocount. The denominator normalizes by the total comparison count n_r and the corresponding

prior terms. The estimate increases monotonically in the fraction of rater r 's judgments that are consistent with the current model, and is given by

$$q_r^{(t)} = \frac{\sum_{i=1}^K \sum_{j=i+1}^K \left[w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)} \right] + (\alpha - 1)}{n_r + \alpha + \beta - 2}, \quad (12)$$

where n_r is the total number of comparisons by rater r .

The update for item skill λ_i is the ratio of two responsibility-weighted quantities. The numerator aggregates item i 's responsibility-weighted wins across all raters, each weighted by γ , and thus quantifies the total evidence favoring item i . The denominator aggregates, over all opponents j , the responsibility-weighted comparisons involving i normalized by the current combined skill $\lambda_i^{(t-1)} + \lambda_j^{(t-1)}$, acting as the effective exposure of item i . Both terms are regularized by the Gamma-prior hyperparameters, giving

$$\lambda_i^{(t)} = \frac{\sum_{r=1}^R \left[\sum_{j=1, j \neq i}^K w_{r,ij} \gamma_{r,ij}^{(t-1)} \right] + (a - 1)}{\sum_{j=1, j \neq i}^K \left[\frac{\sum_{r=1}^R \left[w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)} \right]}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}} \right] + b}. \quad (13)$$

The full derivation of the Expectation-Maximization updates is given in Appendix B.

4 EXPERIMENTAL RESULTS

We evaluate our **Bayesian Bradley-Terry** model with rater **Quality (BBQ)** against two baselines: the Bayesian Bradley-Terry estimator (Bayes-BT) (Caron & Doucet, 2012), and a gradient-based BT estimator with per-rater quality parameters (Crowd-BT) (Chen et al., 2013). For Crowd-BT we use the reference implementation of Google Research (Google, 2025), the same code employed by CLIC (2025) and Ballé et al. (2025). All hyperparameter settings are reported in Section C.

We benchmark the three Bradley-Terry (BT) variants across datasets spanning two domains: human-preference benchmarks for large language models (HUMAINE (Team, 2025), MT-Bench (Zheng et al., 2023)) and for image compression (WD (Ballé et al., 2025), HiFiC (Mentzer et al., 2020), ConHa (Aczel & Wattenhofer, 2024)), together with the newly collected inhomogeneous rater quality (IHQ) dataset derived from the CLIC2024 image test set (CLIC, 2025). To isolate the effect of rater reliability, we partition IHQ into a screened and an unscreened subset. This partition permits a controlled analysis of how low-reliability, high-reliability, and mixed annotator populations affect estimator performance. The IHQ construction protocol is detailed in Section E, and dataset statistics, comparison counts, rater counts, and item counts, are summarized in Section D and Table 2.

Ground-truth rankings are unavailable for the real datasets. As a sanity check, we first verified on simulated data that all three estimators recover the identical ordering in the large-sample limit. For the crowdsourced datasets, we take the ranking estimated on the full dataset as a surrogate ground truth. We assess the validity of this surrogate by inspecting the top-1 item recovered by each estimator. On every dataset except IHQ-unscreened, all three estimators agree on the top-ranked item. On IHQ-unscreened, Crowd-BT and Bayes-BT both fail to rank the reference image highest, whereas BBQ recovers it correctly.

A reliable aggregation estimator should return a stable ordering under repeated sampling of the study. We quantify this stability by bootstrap resampling, which estimates the sampling variability of the recovered rankings. The bootstrap protocol is described in Section F. Note that the WD and HiFiC studies used active, non-i.i.d. selection of comparison pairs: HiFiC applied a binary-search acquisition strategy and WD selected pairs by maximum expected information gain. Because bootstrap resampling assumes exchangeable observations, the bootstrap estimates on these two datasets should be interpreted with caution.

We report two evaluation metrics. Top-1 agreement is the fraction of bootstrap resamples whose top-ranked item coincides with the top-ranked item on the full dataset. Kendall's Tau (τ) measures the ordinal (rank) correlation between two rankings (Kendall, 1938), with values near one indicating high stability. Top-1 agreement is the relevant criterion when only the single best item matters, whereas Kendall's Tau quantifies the stability of the full ordering.

Table 1: Performance of BT-based aggregation estimators across datasets. Top block: Top-1 agreement [%]; bottom block: Kendall’s Tau. Per column, **bold** marks the best value and underline the second best. BBQ most frequently recovers the top-ranked item, attains the highest Kendall’s Tau on a majority of datasets, and ranks second on the remainder.

	HUMAINE	MT-Bench	WD	HiFiC	ConHa	IHQ		
						all	scr.	unscr.
Top-1 Agreement [%]								
Crowd-BT	85.60	100.00	99.29	98.63	<u>66.59</u>	<u>85.31</u>	98.57	<u>33.15</u>
Bayes-BT	<u>97.30</u>	100.00	100.00	100.00	57.30	75.07	<u>98.90</u>	24.32
BBQ (ours)	97.50	100.00	100.00	100.00	77.12	99.32	99.80	61.92
Kendall’s Tau								
Crowd-BT	0.9385	0.9743	0.9279	<u>0.9366</u>	0.9180	<u>0.9245</u>	0.9171	0.8482
Bayes-BT	<u>0.9443</u>	0.9569	0.9459	0.9293	0.9110	0.9240	0.9116	<u>0.8507</u>
BBQ (ours)	0.9463	<u>0.9675</u>	<u>0.9359</u>	0.9525	0.9265	0.9270	<u>0.9132</u>	0.8563

4.1 PERFORMANCE ACROSS DATASETS

We compare the estimators across datasets along two axes: ranking accuracy and ranking stability. The Top-1 agreement and Kendall’s Tau values are reported in Table 1. Datasets with a higher comparison count per model, such as MT-Bench, WD, and HiFiC, are generally easier, since the estimators are less sample-limited. HUMAINE departs from this monotone trend, indicating that estimator performance is governed not only by the aggregate comparison count but also by per-rater consistency and by the diversity of the annotator population.

Across datasets, BBQ attains the strongest stability and robustness overall. It most frequently recovers the top-ranked item, attaining the (tied) highest Top-1 agreement on three datasets. It attains the highest Kendall’s Tau on five of the eight datasets and the second-highest on the remaining three.

The three datasets on which BBQ ranks second in Kendall’s Tau all comprise high-reliability, homogeneous annotator populations. On MT-Bench, all three estimators recover the top item exactly. The WD dataset of Ballé et al. (2025) was labeled by only five raters, presumably the authors, consistent with exceptionally careful annotation. In IHQ-screened, raters were explicitly filtered for quality. When per-rater reliability is uniformly high, the homogeneous-reliability assumption is approximately satisfied, so explicitly modeling rater quality confers little advantage and the margin of BBQ over the simpler estimators narrows.

The IHQ experiments, across both the screened and unscreened subsets, exhibit a consistent pattern. BBQ substantially outperforms Crowd-BT and Bayes-BT precisely when low-reliability raters are present, as in the CLIC-all and CLIC-unscreened conditions. All three estimators attain their highest Top-1 agreement on the screened subset, which quantifies the value of rater pre-screening. Although Crowd-BT also parameterizes rater quality, its Top-1 agreement degrades markedly on the full dataset, plausibly because each crowdsourced rater contributes fewer than 40 comparisons, so its per-rater quality estimates are sample-starved and sensitive to label noise. BBQ, by contrast, retains high performance without screening, incurring only a small drop in Top-1 agreement, which evidences robustness to low-reliability raters.

Attaining uniformly high rater quality is nonetheless difficult in practice. Large-scale crowdsourcing induces reliability heterogeneity, pre-screening is costly, and residual subjective variation affects even diligent annotators. In this regime, BBQ offers a principled estimator that extracts the informative signal from partially reliable raters while down-weighting noisy contributions.

4.2 SCALING WITH RATERS AND COMPARISONS

As Table 1 indicates, a larger comparison count per model generally improves performance. The total comparison count can be increased along two orthogonal axes: by enlarging the rater pool, or by increasing the comparisons elicited per rater.

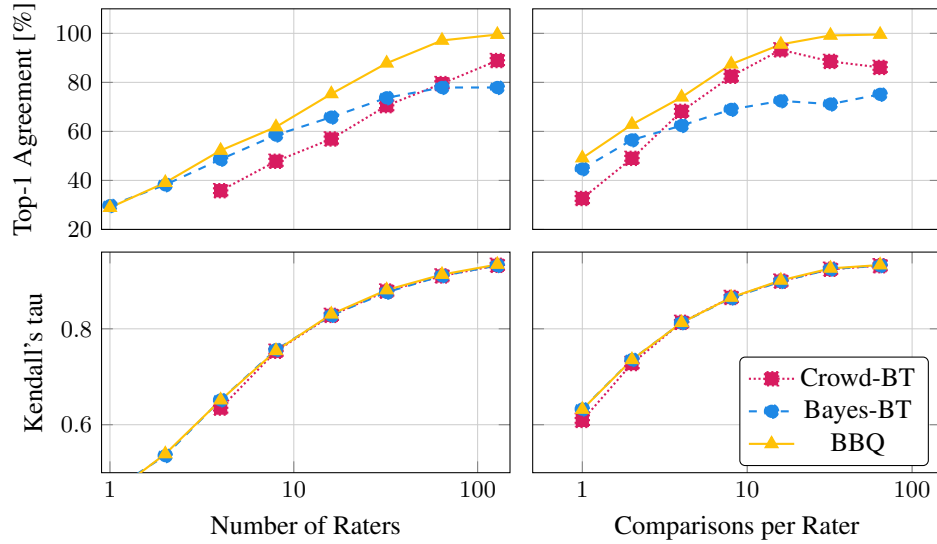


Figure 1: Scaling behavior of the Bradley–Terry variants (Crowd-BT (Caron & Doucet, 2012), Bayes-BT (Chen et al., 2013), and BBQ, ours) on IHQ-all. **Left column:** performance as a function of the number of raters, with comparisons per rater held at its maximum. **Right column:** performance as a function of comparisons per rater, with the number of raters held at its maximum. Both Top-1 agreement and Kendall’s τ increase monotonically with either the number of raters or the number of comparisons. Top-1 agreement separates the estimators, whereas Kendall’s τ is nearly indistinguishable across them. Crowd-BT fails to converge in the few-rater regime, which illustrates the practical benefit of the EM algorithm’s convergence guarantee over gradient ascent. Crowd-BT and BBQ are comparable in the sparse-data regime, but BBQ increasingly dominates Bayes-BT as either axis grows.

Figure 1 disentangles the contribution of each axis for the three BT variants. The left column varies the number of raters with comparisons per rater fixed at its maximum, and the right column varies comparisons per rater with the rater pool fixed at its maximum. The estimators separate clearly in Top-1 agreement while remaining nearly indistinguishable in Kendall’s τ . Crowd-BT fails to converge with only one or two raters, which underscores the advantage of EM, whose convergence guarantee the gradient-based baseline lacks.

In the small-sample regime, whether few raters or few comparisons per rater, Crowd-BT and BBQ perform comparably. BBQ requires both a sufficient rater pool and sufficient comparisons per rater to identify the per-rater quality parameters, since these parameters are otherwise weakly identified. As either axis grows, BBQ increasingly outperforms Bayes-BT. Bayes-BT is the weakest estimator overall, but surpasses Crowd-BT under limited data per rater or a small rater pool, precisely the regime in which Crowd-BT cannot reliably estimate rater quality.

4.3 RATER QUALITY

Figure 2 relates each rater’s agreement with the full-dataset ranking to the rater-quality parameter q_r inferred by BBQ. The two quantities are positively correlated: raters whose judgments align more closely with the consensus ranking are assigned higher inferred quality, confirming that q_r is an interpretable reliability estimate. The Pearson correlation is $r = 0.551$ over the 50 raters of the screened subset and a stronger $r = 0.724$ over the 62 raters of the unscreened subset.

As expected, the unscreened subset contains several raters exhibiting jointly low agreement and low inferred quality. A subset of the IHQ-unscreened raters fall below the chance level of 50% agreement, i.e., they systematically disagree with the majority. This regime is precisely where modeling rater reliability is essential for correct aggregation. BBQ identifies these raters and assigns them low q_r , which down-weights their contribution to the likelihood and attenuates the label noise they introduce into the recovered ranking.

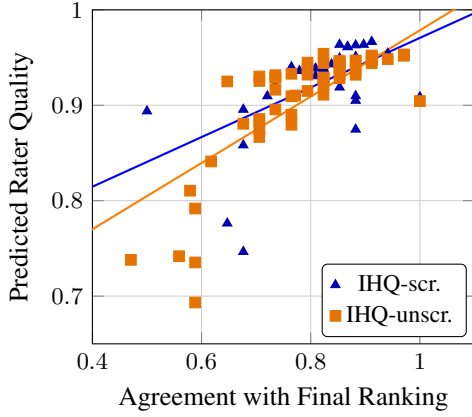


Figure 2: Per-rater agreement with the full-dataset ranking (abscissa) against the BBQ-inferred rater quality q_r (ordinate) for the IHQ datasets. Each marker is an individual rater; triangles denote the screened (filtered) subset and squares the unscreened (unfiltered) subset. Overlaid lines are ordinary least-squares fits with the reported slope, intercept, and Pearson correlation r .

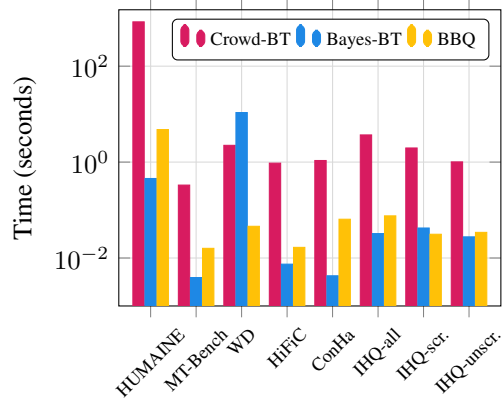


Figure 3: Mean wall-clock time in seconds (log scale) to fit each estimator on a single bootstrap resample, across the eight datasets. Crowd-BT incurs substantial runtime on some datasets, whereas BBQ remains uniformly fast.

4.4 COMPUTATIONAL EFFICIENCY

Figure 3 reports the mean wall-clock time to fit each estimator on a single bootstrap resample, per dataset. These measurements characterize the computational feasibility of each estimator in practical evaluation pipelines. BBQ converges within a few seconds on every dataset. Crowd-BT is uniformly slower than BBQ, with pronounced runtime on high-comparison-count datasets such as HUMANE, where a single fit requires roughly 15 minutes. Bayes-BT converges slowly on WD but is marginally faster than BBQ on the remaining datasets. The efficiency of BBQ follows directly from its closed-form EM updates, whose per-iteration cost is low and which converge in few iterations even on large datasets.

We note further that the Crowd-BT baseline is a heavily optimized C implementation (Kernighan & Ritchie, 1988), whereas BBQ is implemented in plain NumPy without low-level optimization. The reported runtimes are therefore a conservative upper bound on BBQ’s achievable latency; a compiled or fully vectorized implementation would reduce it further. Consequently, BBQ is not only more robust and stable but also computationally practical for large-scale human-preference studies and for pipelines that require repeated bootstrap resampling. The combination of ranking accuracy, stability, and low runtime makes BBQ well suited to production deployment.

4.5 CONFIDENCE INTERVALS

We assess the calibration of the uncertainty estimates produced by Crowd-BT, Bayes-BT, and BBQ. In a controlled simulation under the null hypothesis H_0 that two items are equally skilled, raters supply pairwise comparisons generated by fair coin flips. Each estimator fits the item skills and constructs an interval estimate, and H_0 is rejected when the two intervals are disjoint (Section G). The resulting empirical Type I error rates are shown in Figure 6. At the 99% confidence level the nominal Type I error rate is 1%. Crowd-BT and BBQ both track this nominal rate closely, at slightly below 1%, whereas Bayes-BT is markedly conservative at $\sim 0.1\%$. Hence Crowd-BT and BBQ yield well-calibrated interval estimates, while Bayes-BT systematically under-rejects and thus underestimates the false-positive rate.

5 LIMITATIONS

While BBQ models rater quality effectively, its advantage over simpler estimators shrinks when annotator reliability is homogeneous. On such datasets the mixture component modeling unreliable raters is superfluous and yields little additional benefit. Enforcing uniformly high reliability, however, entails substantial screening cost and effort, which is often infeasible at scale. This trade-off implies that BBQ is most beneficial in realistic crowdsourced regimes and least beneficial when evaluations are tightly curated.

A second limitation concerns the admissible observation model. BBQ is currently specified for binary pairwise comparisons, whereas many evaluation studies elicit graded ratings, full rankings, or multi-way (k -alternative) choices. Extending the likelihood to accommodate these response formats would broaden the estimator’s applicability.

Methodologically, our stability estimates are computed from a finite number of bootstrap resamples. This yields a practical estimate of ranking variability, but tighter validation would require larger resample counts and larger studies. Moreover, the likelihood assumes conditional independence across comparisons and therefore does not capture contextual or order effects that are known to bias human judgments.

Finally, although BBQ scales favorably, very large item or rater counts may still be computationally demanding absent an optimized or compiled implementation.

6 CONCLUSION

We introduced BBQ, a Bayesian Bradley-Terry model that jointly infers item skills and per-rater reliability. Its central contribution is an Expectation-Maximization (EM) algorithm with closed-form updates that estimates the item skills and the rater-quality parameters simultaneously, thereby down-weighting or excluding unreliable annotators. Modeling rater quality explicitly yields more stable and more accurate rankings. The EM algorithm converges rapidly and is guaranteed to increase the observed-data likelihood monotonically, removing the convergence pathologies of gradient-based estimators.

Across the evaluated datasets, BBQ attained high Top-1 agreement and the strongest Kendall’s Tau, evidencing both robustness and reliability. It is most advantageous under noisy or crowdsourced annotation, retaining accuracy even when each rater contributes few comparisons. This makes BBQ particularly suited to large-scale settings in which annotators are predominantly non-experts and vary widely in attentiveness. Our results indicate that modeling rater quality is decisive precisely when annotator reliability is heterogeneous or partially unreliable. Because the estimator is Bayesian, it additionally returns principled posterior uncertainty over item scores, which supports interpretable cross-study comparison. We showed that the resulting interval estimates are well-calibrated and can serve as significance tests for differences between items. Finally, the inferred rater-quality parameters correlate with agreement to the consensus ranking, validating that the estimator recovers annotator reliability.

In sum, BBQ advances human-evaluation methodology by providing a practical, interpretable, and broadly applicable estimator for aggregating noisy pairwise comparisons, and thereby contributes toward more cost-efficient, interpretable, and reproducible human studies of AI-generated content.

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A USAGE OF LLMs

In preparing this manuscript, we used large language models (LLMs) strictly as auxiliary tools. ChatGPT, Claude, Gemini, and Grammarly were used for spellchecking, phrasing refinement, and text condensation to improve clarity and readability. ChatGPT, Claude, and Cursor were additionally used for code analysis, code completion, and visualization generation within our development workflow. These tools served solely as auxiliary aids; all core research contributions, experimental design, and interpretation of results were developed independently by the authors.

B BAYESIAN BT WITH RATER QUALITY DERIVATION

We consider R raters comparing K items. Rater r has reliability q_r : with probability q_r the rater draws a judgment from the Bradley–Terry model, and with probability $1 - q_r$ the rater responds uninformatively (a fair coin flip). The following is a standard EM (Dempster et al., 1977) derivation that embeds these per-rater reliabilities within the Bayesian estimation framework.

B.1 MODEL DEFINITION

Notation:

- $\lambda = (\lambda_1, \dots, \lambda_K)$: skill parameters for the K items.
- $q = (q_1, \dots, q_R)$: quality parameters for the R raters.
- $Y_i \sim \mathcal{E}(\lambda_i)$: latent arrival time associated with item i .
- $A_{r,ij}^{(c)} \sim \text{Bernoulli}(q_r)$: indicator that the c -th comparison of pair (i, j) by rater r follows the Bradley–Terry model.
- $n_{r,ij}$: total number of comparisons of (i, j) by rater r .
- $n_{ij} = \sum_{r=1}^R n_{r,ij}$: total number of comparisons of (i, j) .
- $w_{r,ij}$: number of times rater r ranked i above j .
- $w_{ij} = \sum_{r=1}^R w_{r,ij}$: total number of times i was ranked above j .
- $m_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)}$: number of Bradley–Terry-model comparisons of (i, j) by rater r .
- $m_{ij} = \sum_{r=1}^R m_{r,ij}$: total number of Bradley–Terry-model comparisons of (i, j) .
- $v_{r,ij} = \sum_{c=1}^{w_{r,ij}} A_{r,ij}^{(c)}$: number of Bradley–Terry-model wins of i over j by rater r .
- $v_{ij} = \sum_{r=1}^R v_{r,ij}$: total number of Bradley–Terry-model wins of i over j .
- $Z_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)} \min(Y_i^{(c)}, Y_j^{(c)})$: sum of minimal arrival times, with

$$Z_{r,ij} \mid m_{r,ij} \sim \Gamma(m_{r,ij}, \lambda_i + \lambda_j).$$

Bradley–Terry probability:

$$P(i \text{ beats } j) = \frac{\lambda_i}{\lambda_i + \lambda_j}.$$

Mixture with rater quality:

$$P(r \text{ ranks } i \text{ above } j) = q_r \frac{\lambda_i}{\lambda_i + \lambda_j} + (1 - q_r) \frac{1}{2}.$$

Latent exponential view:

$$Y_i \sim \mathcal{E}(\lambda_i), \quad Y_j \sim \mathcal{E}(\lambda_j), \quad P(i \text{ beats } j) = P(Y_i < Y_j).$$

B.2 COMPLETE-DATA LOG-LIKELIHOOD

We need to compute:

$$\begin{aligned}\ell_c(\lambda, q; D, Z, A) &= \log P(D, Z, A \mid \lambda, q) \\ &= \log P(Z \mid D, A, \lambda, q) + \log P(A \mid D, \lambda, q) + \log P(D \mid \lambda, q)\end{aligned}$$

Log-Likelihood of $P(Z \mid A, \lambda)$:

$$\begin{aligned}\log P(Z \mid D, A, \lambda, q) &= \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{ij} \log(\lambda_i + \lambda_j) - (\lambda_i + \lambda_j) z_{ij} \right. \\ &\quad \left. + (m_{ij} - 1) \log z_{ij} - \log \Gamma(m_{ij}) \right]\end{aligned}$$

Log-Likelihood of $P(A \mid q)$:

$$\log P(A \mid D, \lambda, q) = \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{r,ij} \log q_r + (n_{r,ij} - m_{r,ij}) \log(1 - q_r) \right]$$

Log-Likelihood of $P(D \mid \lambda, A)$:

$$\begin{aligned}\log P(D \mid \lambda, q) &= \sum_{r=1}^R \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K \left[v_{r,ij} \log \frac{\lambda_i}{\lambda_i + \lambda_j} + (w_{r,ij} - v_{r,ij}) \log \frac{1}{2} \right] \\ &= \sum_{r=1}^R \sum_{i=1}^K [v_{ir} \log \lambda_i] \\ &\quad - \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K [(v_{r,ij} + v_{r,ji}) \log(\lambda_i + \lambda_j) + (w_{r,ij} + w_{r,ji} - v_{r,ij} - v_{r,ji}) \log 2] \\ &= \sum_{i=1}^K [v_i \log \lambda_i] \\ &\quad - \sum_{i=1}^K \sum_{j=i+1}^K [m_{ij} \log(\lambda_i + \lambda_j) + (n_{ij} - m_{ij}) \log 2]\end{aligned}$$

Complete-Data Log-Likelihood:

$$\begin{aligned}
\ell_c(\lambda, q; D, Z, A) &= \log P(Z \mid D, A, \lambda, q) + \log P(A \mid D, \lambda, q) + \log P(D \mid \lambda, q) \\
&= \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{ij} \log(\lambda_i + \lambda_j) - (\lambda_i + \lambda_j) z_{ij} + (m_{ij} - 1) \log z_{ij} - \log \Gamma(m_{ij}) \right] \\
&\quad + \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{r,ij} \log q_r + (n_{r,ij} - m_{r,ij}) \log(1 - q_r) \right] \\
&\quad + \sum_{i=1}^K [v_i \log \lambda_i] \\
&\quad - \sum_{i=1}^K \sum_{j=i+1}^K [m_{ij} \log(\lambda_i + \lambda_j) + (n_{ij} - m_{ij}) \log 2] \\
&= \sum_{i=1}^K \sum_{j=i+1}^K \left[(n_{ij} - m_{ij}) \log 2 - (\lambda_i + \lambda_j) z_{ij} + (m_{ij} - 1) \log z_{ij} - \log \Gamma(m_{ij}) \right] \\
&\quad + \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{r,ij} \log q_r + (n_{r,ij} - m_{r,ij}) \log(1 - q_r) \right] \\
&\quad + \sum_{i=1}^K [v_i \log \lambda_i]
\end{aligned}$$

B.3 EXPECTATION STEP

We place conjugate priors on the parameters: a Gamma prior $\lambda_i \sim \Gamma(a, b)$ on each item skill i , and a Beta prior $q_r \sim B(\alpha, \beta)$ on each rater reliability r .

The Q -function is the expectation of the complete-data log-posterior, taken under the latent-variable posterior at the current estimates:

$$Q(\lambda, q \mid \lambda^*, q^*) = \mathbb{E}_{A, Z \mid D, \lambda^*, q^*} [\ell_c(\lambda, q; D, Z, A) + \log P(\lambda) + \log P(q)].$$

Posterior probability of a Bradley-Terry-consistent annotation:

$$\begin{aligned}
P(A_{r,ij}^{(k)} = 1 \mid i \succ j) &= \frac{P(i \succ j \mid A = 1) P(A = 1)}{P(i \succ j \mid A = 1) P(A = 1) + P(i \succ j \mid A = 0) P(A = 0)} \\
&= \frac{\left(\frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*} \right) q_r^*}{\left(\frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*} \right) q_r^* + \frac{1}{2} (1 - q_r^*)} = \gamma_{r,ij}^*.
\end{aligned}$$

Expected sufficient statistics:

$$\begin{aligned}
\mathbb{E}[m_{r,ij}] &= w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^* \\
\mathbb{E}[m_{ij}] &= \sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \\
\mathbb{E}[v_{r,ij}] &= w_{r,ij} \gamma_{r,ij}^* \\
\mathbb{E}[v_{ir}] &= \sum_{j, j \neq i}^K w_{r,ij} \gamma_{r,ij}^* \\
\mathbb{E}[v_i] &= \sum_{r=1}^R \sum_{j, j \neq i}^K w_{r,ij} \gamma_{r,ij}^* \\
\mathbb{E}[z_{ij}] &= \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*}.
\end{aligned}$$

Expected complete-data log-posterior:

$$\begin{aligned}
\mathbb{E}[\ell_c] &= \sum_{i=1}^K \sum_{j=i+1}^K \left[-(\lambda_i + \lambda_j) \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} \right] \\
&+ \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[(w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \log q_r \right. \\
&\quad \left. + (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) \log(1 - q_r) \right] \\
&+ \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K w_{r,ij} \gamma_{r,ij}^* \log \lambda_i \\
&+ \text{const.}
\end{aligned}$$

Priors contribute:

$$\begin{aligned}
\mathbb{E}[\log P(\lambda)] &= \sum_{i=1}^K [(a-1) \log \lambda_i - b \lambda_i], \\
\mathbb{E}[\log P(q)] &= \sum_{r=1}^R [(\alpha-1) \log q_r + (\beta-1) \log(1 - q_r)].
\end{aligned}$$

Final Q -function:

$$\begin{aligned}
Q(\lambda, q \mid \lambda^*, q^*) = & \sum_{i=1}^K \sum_{j=i+1}^K \left[-(\lambda_i + \lambda_j) \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} \right] \\
& + \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[(w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \log q_r \right. \\
& \quad \left. + (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) \log(1 - q_r) \right] \\
& + \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K w_{r,ij} \gamma_{r,ij}^* \log \lambda_i \\
& + \sum_{i=1}^K [(a - 1) \log \lambda_i - b \lambda_i] \\
& + \sum_{r=1}^R [(\alpha - 1) \log q_r + (\beta - 1) \log(1 - q_r)] \\
& + \text{const.}
\end{aligned}$$

where

$$\gamma_{r,ij}^* = \frac{q_r^* \frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*}}{q_r^* \frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*} + (1 - q_r^*) \frac{1}{2}}.$$

B.4 M-STEP

The M-step maximizes the Q -function with respect to the parameters (λ, q) , holding fixed the expected sufficient statistics computed in the E-step.

Update for q_r The update for each rater reliability q_r is obtained by setting the partial derivative of Q (including the Beta-prior contribution) with respect to q_r to zero.

$$\begin{aligned}
Q(q \mid \lambda^*, q^*) &= \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[(w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \log q_r \right. \\
&\quad \left. + (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) \log(1 - q_r) \right] \\
&\quad + (\alpha - 1) \log q_r + (\beta - 1) \log(1 - q_r) + \text{const.} \\
\frac{\partial Q}{\partial q_r} &= \sum_{i=1}^K \sum_{j=i+1}^K \left[\frac{w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*}{q_r} - \frac{n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{1 - q_r} \right] \\
&\quad + \frac{\alpha - 1}{q_r} - \frac{\beta - 1}{1 - q_r} \\
0 &\stackrel{!}{=} \frac{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) + (\alpha - 1)}{q_r} \\
&\quad - \frac{\sum_{i=1}^K \sum_{j=i+1}^K (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) + (\beta - 1)}{1 - q_r} \\
\Rightarrow &\quad (1 - q_r) \left(\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) + (\alpha - 1) \right) \\
\Rightarrow &\quad q_r \left(\sum_{i=1}^K \sum_{j=i+1}^K (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) + (\beta - 1) \right) \\
\Rightarrow &\quad q_r = \frac{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) + (\alpha - 1)}{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^* + n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) + (\beta + \alpha - 2)} \\
\Rightarrow &\quad q_r^{(t)} = \frac{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)}) + (\alpha - 1)}{n_r + \beta + \alpha - 2}
\end{aligned}$$

where the last line gives the explicit update at iteration t .

Update for λ_i The update for each item skill λ_i is obtained by setting the partial derivative of Q (including the Gamma-prior contribution) with respect to λ_i to zero.

$$\begin{aligned}
Q(\lambda \mid \lambda^*, q^*) &= \sum_{i=1}^K \sum_{j=i+1}^K -(\lambda_i + \lambda_j) \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} \\
&\quad + \sum_{r=1}^R \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K w_{r,ij} \gamma_{r,ij}^* \log \lambda_i \\
&\quad + \sum_{i=1}^K [(a - 1) \log \lambda_i - b \lambda_i] + \text{const.}
\end{aligned}$$

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$$\begin{aligned}
\frac{\partial Q}{\partial \lambda_i} &= - \sum_{j \neq i}^K \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} + \sum_{r=1}^R \frac{\sum_{j \neq i}^K w_{r,ij} \gamma_{r,ij}^*}{\lambda_i} + \frac{a-1}{\lambda_i} - b \\
0 &\stackrel{!}{=} \frac{\sum_{r=1}^R \sum_{j \neq i}^K w_{r,ij} \gamma_{r,ij}^* + (a-1)}{\lambda_i} - \sum_{j \neq i}^K \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} - b \\
\Rightarrow \lambda_i^{(t)} &= \frac{\sum_{r=1}^R \sum_{j \neq i}^K w_{r,ij} \gamma_{r,ij}^{(t-1)} + (a-1)}{\sum_{j \neq i}^K \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)})}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}} + b} \\
\gamma_{r,ij}^{(t-1)} &= \frac{q_r^{(t-1)} \frac{\lambda_i^{(t-1)}}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}}}{q_r^{(t-1)} \frac{\lambda_i^{(t-1)}}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}} + (1 - q_r^{(t-1)})^{\frac{1}{2}}}
\end{aligned}$$

where the last line gives the explicit update at iteration t .

C HYPERPARAMETERS

For Crowd-BT, we use the implementation of Google (2025) with its default hyperparameters.

The only hyperparameters of the Bayes-BT and BBQ estimators are the prior parameters and the convergence threshold. For both estimators, we declare convergence when the maximum change in any ELO score between successive iterations falls below 1. The ELO score is a monotone transform of the skill parameter λ ,

$$\text{ELO} = \log(\text{skill}) \cdot \text{ELO_SCALE_FACTOR}. \quad (14)$$

where the ELO_SCALE_FACTOR is fixed at 400.

For both Bayes-BT and BBQ we place a Gamma prior with shape $a = 5$ and rate $b = 0.1$ on the item skills. For BBQ, the Beta prior on rater quality uses $\alpha = 10$ and $\beta = 2$.

D DATASETS

We evaluate the estimators on a diverse collection of human-preference datasets spanning the language and image domains. Table 2 summarizes their statistics.

For natural language, we use the HUMAINE dataset (Team, 2025), a large-scale benchmark comprising over 100k comparisons, and MT-Bench (Zheng et al., 2023), which supplies crowd-worker model-comparison judgments on multi-turn dialogues.

For image compression, we use three datasets: WD (Ballé et al., 2025), a dense expert-labeled dataset with thousands of comparisons per rater; HiFiC (Mentzer et al., 2020), which evaluates learned image codecs; and ConHa (Aczel & Wattenhofer, 2024), which targets conditional generative models. These datasets differ substantially in scale, rater count, and rater expertise, and thus constitute a broad testbed for robustness.

Finally, we introduce the inhomogeneous rater quality (**IHQ**) dataset, collected in a user study on the CLIC2024 (CLIC, 2025) data conducted on the *Mabyduck* platform. It comprises two-alternative forced-choice (2AFC) judgments over 28 generative image-compression models. To isolate the effect of rater reliability, we provide two subsets: (i) a **screened** subset, containing only raters who passed pre-screening checks for attention and display quality, and (ii) an **unscreened** subset, containing all raters. The screened subset represents a high-reliability annotator population, while the unscreened subset reproduces the label-noise conditions characteristic of large-scale human evaluation.

Table 2: Summary statistics of the datasets used in our experiments. The IHQ dataset was collected by us on the Mabyduck platform and includes both 2AFC and 3AFC settings. For IHQ we report screened and unscreened subsets: the screened subset retains only raters who passed pre-screening tests for attention and display quality, yielding higher reliability, whereas the unscreened subset retains all raters.

dataset	# comparisons	# raters	# models	comp/rater	comp/model
HUMAINE (Team, 2025)	105,220	1,977	27	53.2	3897.0
MT-Bench (Zheng et al., 2023)	3,355	65	6	51.6	559.2
WD (Ballé et al., 2025)	16,659	5	30	3331.8	555.3
HiFiC (Mentzer et al., 2020)	5,220	20	9	261.0	580.0
ConHa (Aczel & Wattenhofer, 2024)	1,531	40	8	38.3	191.4
IHQ-all	4,074	112	28	36.4	145.5
IHQ-screened	2,012	50	28	40.2	71.9
IHQ-unscreened	2,062	62	28	33.3	73.6

Together, these datasets span both large-scale, comparatively clean benchmarks and smaller, noisier regimes in which modeling rater reliability is decisive.

E USER STUDY PLATFORM

All pairwise comparisons on the CLIC2024 (CLIC, 2025) dataset were collected on the Mabyduck platform (Ltd., 2025). Figure 4 shows the annotation interface. Each trial poses a fixed 2AFC prompt, “Which image looks more similar to the reference image?”; ties are disallowed, and comparison pairs were sampled uniformly at random.

To study rater reliability, we partition the IHQ dataset into screened and unscreened subsets. Raters in the screened subset passed pre-tests for color vision, display contrast, and sensitivity to subtle differences, and additionally completed training comparisons, in which they selected the higher-quality image, to calibrate them to the evaluation criteria.

Figure 5 presents four representative stimuli from the pre-screening protocol. The first two are standard color-blindness tests, in which the rater must correctly identify the numeral embedded in each pattern. The final two are low-contrast shape-detection tests that probe sensitivity to subtle contrast differences: one presents a light-gray shape on a white background, the other a dark-gray shape on a black background. These tests gate admission to the screened subset, ensuring that only raters with adequate visual acuity contribute to it.

F BOOTSTRAPPING DETAILS

Because BBQ and Crowd-BT both parameterize rater quality, we bootstrap over raters rather than over individual comparisons. Resampling at the rater level preserves each rater’s empirical comparison distribution and yields an unbiased assessment of ranking stability. We draw 10,000 bootstrap resamples of raters for every dataset, except HUMAINE, where a single Crowd-BT fit requires approximately 15 minutes (see Section 4.4). For HUMAINE we therefore reduce the count to 1,000 resamples to bound total computation.

G UNCERTAINTY ESTIMATION DETAILS

Beyond producing a point ranking, uncertainty estimation is essential for assessing whether observed score differences are statistically significant.

The Bayes-BT and BBQ estimators quantify uncertainty through the posterior distribution over item skills. Each skill carries a Gamma prior that is updated by conditioning on the observed pairwise comparisons, and the resulting posterior credible intervals are mapped onto the Elo scale to make them comparable across items.

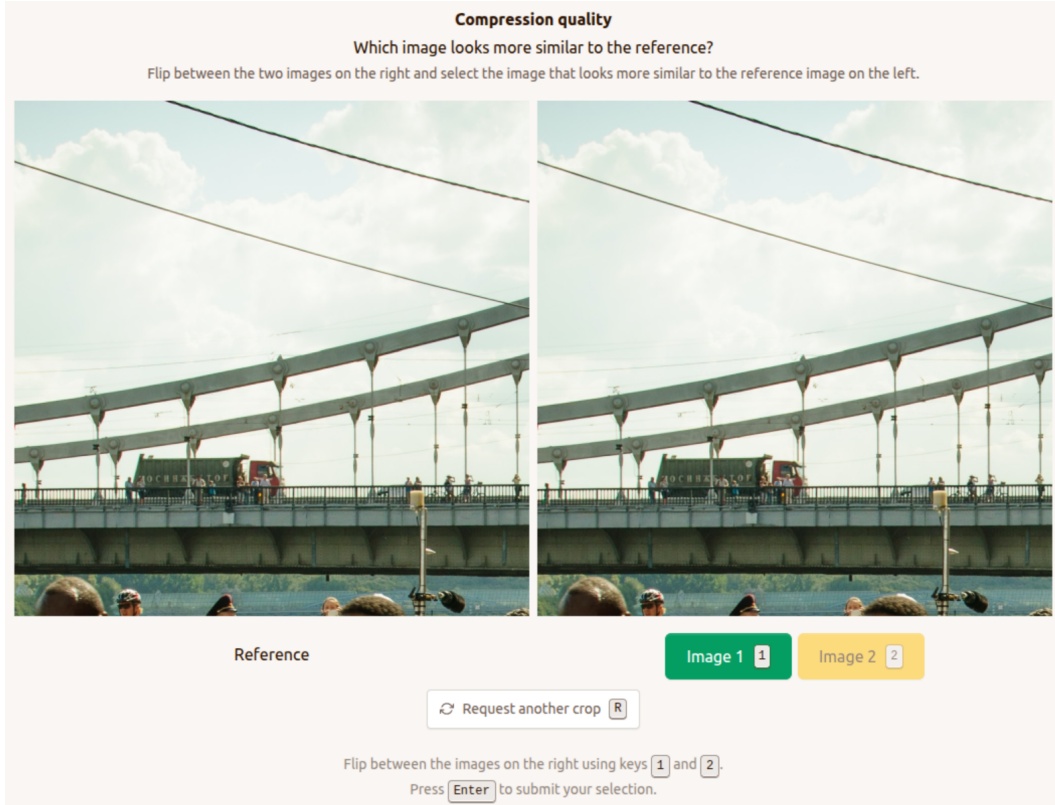


Figure 4: Annotation interface of the Mabyduck user-study platform used to collect the pairwise comparisons. The reference image is displayed on the left, and the rater selects between the two candidate compressed images on the right.

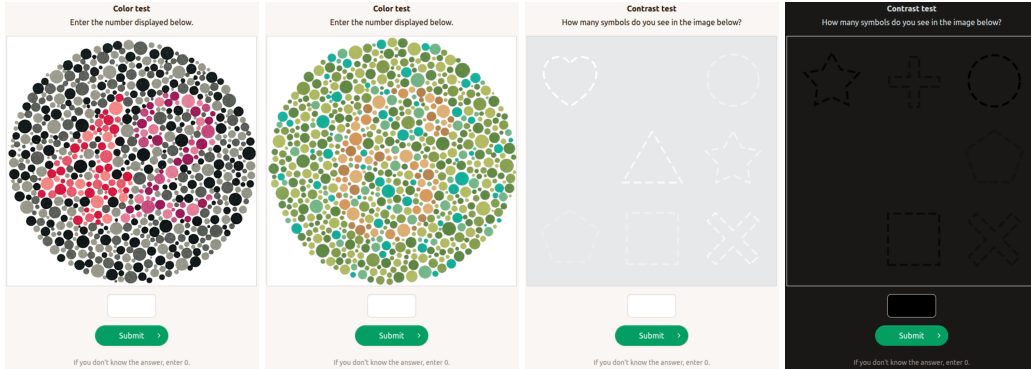


Figure 5: Four representative stimuli from the rater pre-screening protocol. The first two are color-blindness tests, in which the rater must identify the numeral embedded in each pattern. The last two are low-contrast shape-detection tests, comprising a light-gray shape on a white background and a dark-gray shape on a black background, respectively.

Crowd-BT, being non-Bayesian, estimates uncertainty by a frequentist Laplace approximation. Specifically, it forms a second-order Taylor expansion of the log-likelihood about the maximum-likelihood estimate. Inverting the Hessian of the log-likelihood yields the asymptotic covariance matrix, whose diagonal entries are the per-item variances. These variances are converted to 99% confidence intervals via

$$p_{99} = \sqrt{\text{diag}(\text{covariance})} \times k_{\text{Erfc}0.01} \times \sqrt{2} \approx \sqrt{\text{diag}(\text{covariance})} \times 3.29,$$

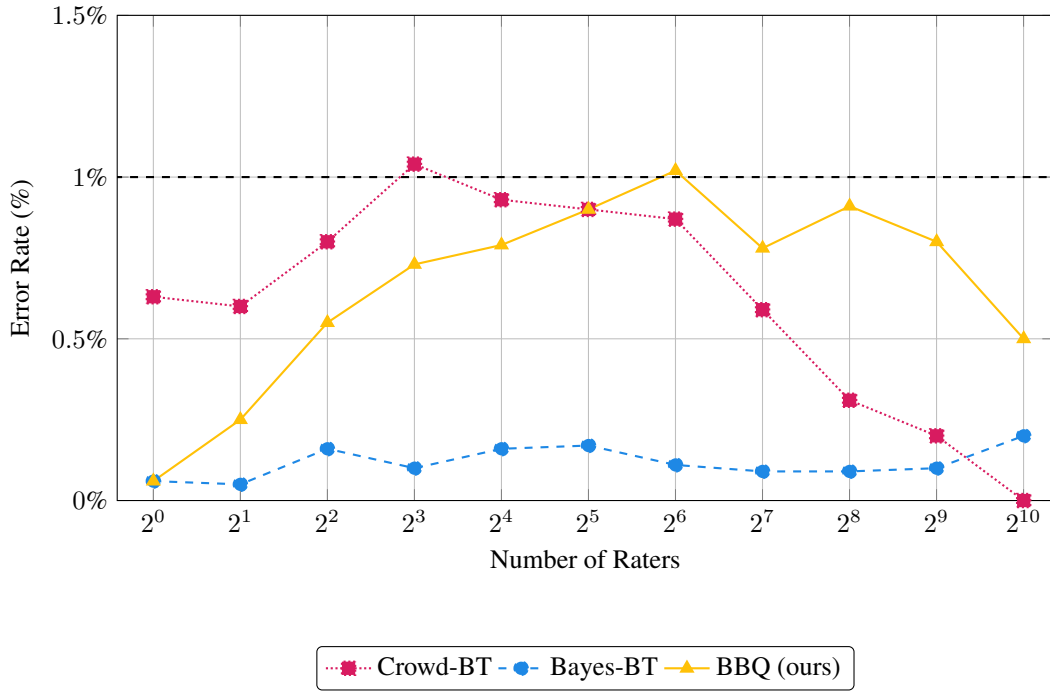


Figure 6: Empirical Type I error rate as a function of the number of raters. The dashed line marks the nominal rate of 1% at the 99% confidence level.

where the multiplicative constant scales the estimated standard deviation to the 99% confidence level. A sharply peaked likelihood yields a small covariance and hence a narrow interval (high certainty), whereas a flat likelihood yields a wide interval (high uncertainty).

To assess the calibration of these uncertainty estimates, we run a controlled simulation. We generate trials in which two items have equal skill (a 50–50 win probability) and synthesize pairwise comparisons by fair coin flips. Each trial simulates multiple raters, and each estimator is fit to recover the Elo scores and their associated interval estimates.

Formally, the null hypothesis H_0 asserts that the two items are equally skilled. For each trial we compute the 99% confidence (or credible) interval of each item’s skill estimate. An estimator falsely rejects H_0 when the two intervals are disjoint, i.e., it declares a significant difference where none exists. The metric of interest is the frequency of such false rejections, i.e., the empirical Type I error rate.

We run 10,000 trials with 50 comparisons per rater, sweeping the number of raters to characterize how the uncertainty estimates scale with sample size. This yields an empirical Type I error rate per estimator, which we compare against the nominal 1% implied by the 99% confidence level. As shown in Figure 6, Crowd-BT and BBQ attain Type I error rates near the nominal 1%, indicating well-calibrated interval estimates. Bayes-BT, by contrast, is systematically conservative, producing error rates well below nominal, which indicates that its credible intervals are miscalibrated for significance testing.